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# Attention Mechanisms in Computer Vision: A Survey

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**Abstract**—Humans can naturally and effectively find salient regions in complex scenes. Motivated by this observation, attention mechanisms were introduced into computer vision with the aim of imitating this aspect of the human visual system. Such an attention mechanism can be regarded as a dynamic weight adjustment process based on features of the input image. Attention mechanisms have achieved great success in many visual tasks, including image classification, object detection, semantic segmentation, video understanding, image generation, 3D vision, multi-modal tasks and self-supervised learning. In this survey, we provide a comprehensive review of various attention mechanisms in computer vision and categorize them according to approach, such as channel attention, spatial attention, temporal attention and branch attention; a related repository https://github.com/MenghaoGuo/Awesome-Vision-Attentions is dedicated to collecting related work. We also suggest future directions for attention mechanism research.

**Index Terms**—Attention, Transformer, Survey, Computer Vision, Deep Learning, Salience.

## 1 Introduction

Methods for diverting attention to the most important regions of an image and disregarding irrelevant parts are called attention mechanisms; the human visual system uses one [1], [2], [3], [4] to assist in analyzing and understanding complex scenes efficiently and effectively. This in turn has inspired researchers to introduce attention mechanisms into computer vision systems to improve their performance. In a vision system, an attention mechanism can be treated as a dynamic selection process that is realized by adaptively weighting features according to the importance of the input. Attention mechanisms have provided benefits in very many visual tasks, e.g. image classification [5], [6], object detection [7], [8], semantic segmentation [9], [10], face recognition [11], [12], person re-identification [13], [14], action recognition [15], [16], few-show learning [17], [18], medical image processing [19], [20], image generation [21], [22], pose estimation [23], super resolution [24], [25], 3D vision [26], [27], and multi-modal task [28], [29].

In the past decade, the attention mechanism has played an increasingly important role in computer vision; Fig. 3, briefly summarizes the history of attention-based models in computer vision in the deep learning era. Progress can be coarsely divided into four phases. The first phase begins from RAM [31], pioneering work that combined deep neural networks with attention mechanisms. It recurrently predicts the important region and updates the whole network in an end-to-end manner through a policy gradient. Later, various works [21], [35] adopted a similar strategy for attention in vision. In this phase, recurrent neural networks(RNNs) were necessary tools for an attention mechanism. At the start of the second phase, Jaderberg et al. [32] proposed the STN which introduces a sub-network to predict an affine transformation used to to select important regions in the input. Explicitly predicting discriminatory input features is the major characteristic of the second phase; DCNs [7], [36] are representative works. The third phase began with SENet [5]

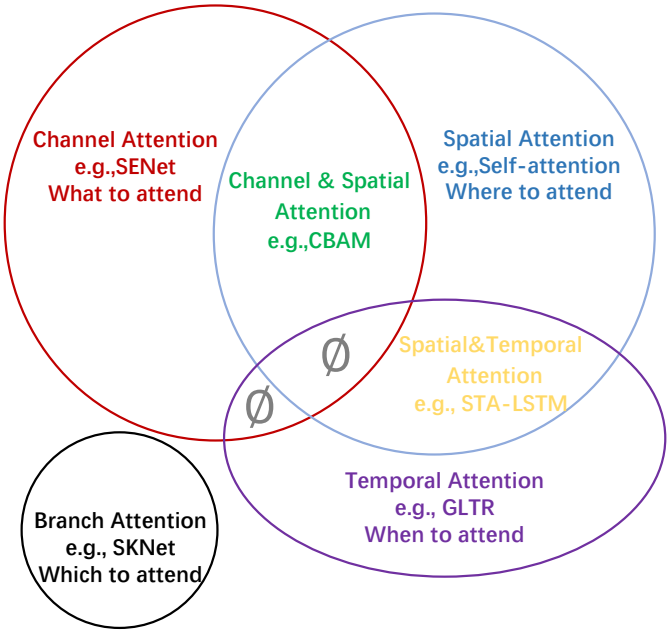


Fig. 1. Attention mechanisms can be categorised according to data domain. These include four fundamental categories of channel attention, spatial attention, temporal attention and branch attention, and two hybrid categories, combining channel & spatial attention and spatial & temporal attention. ∅ means such combinations do not (yet) exist.

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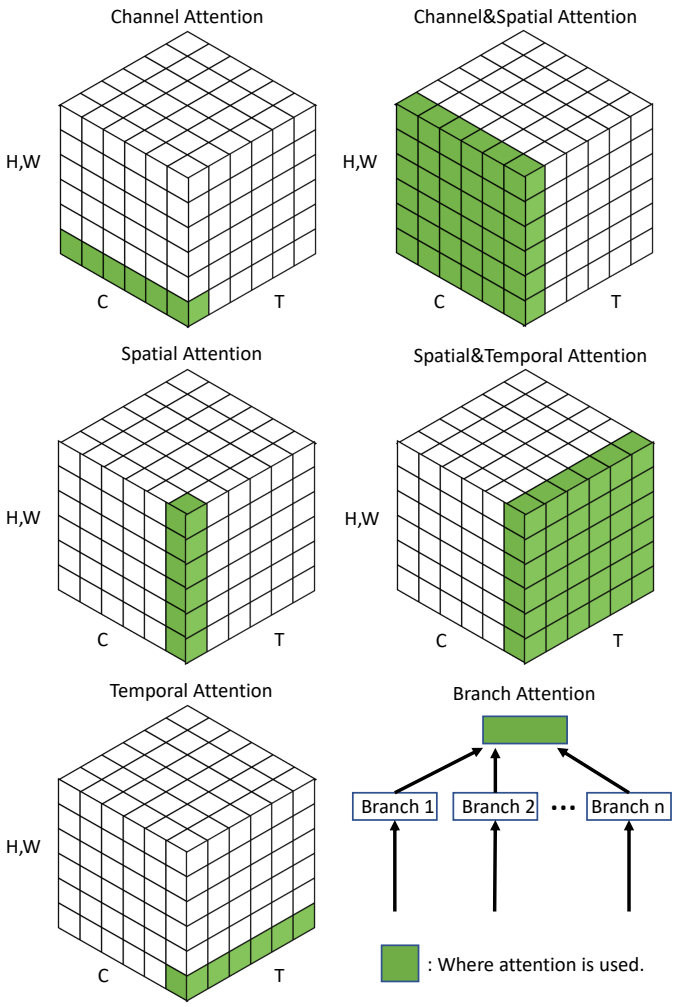


Fig. 2. Channel, spatial and temporal attention can be regarded as operating on different domains. $C$ represents the channel domain, $H$ and $W$ represent spatial domains, and $T$ means the temporal domain. Branch attention is complementary to these. Figure following [30].

that presented a novel channel-attention network which implicitly and adaptively predicts the potential key features. CBAM [6] and ECANet [37] are representative works of this phase. The last phase is the self-attention era. Self-attention was firstly proposed in [33] and rapidly provided great advances in the field of natural language processing [33], [38], [39]. Wang et al. [15] took the lead in introducing self-attention to computer vision and presented a novel non-local network with great success in video understanding and object detection. It was followed by a series of works such as EMANet [40], CCNet [41], HamNet [42] and the Stand-Alone Network [43], which improved speed, quality of results, and generalization capability. Recently, various pure deep self-attention networks (visual transformers) [27], [34], [44], [45], [46], [47], [48], [49] have appeared, showing the huge potential of attention-based models. It is clear that attention-based models have the potential to replace convolutional neural networks and become a more powerful and general architecture in computer vision.

The goal of this paper is to summarize and classify current attention methods in computer vision. Our approach

TABLE 1
Key notation in this paper. Other minor notation is explained where used.

| Symbol | Description |
|---|---|
| $X$ | input feature map, $X \in \mathbb{R}^{C\times H\times W}$ |
| $Y$ | output feature map |
| $W$ | learnable kernel weight |
| FC | fully-connected layer |
| Conv | convolution |
| GAP | global average pooling |
| GMP | global max pooling |
| $[\ ]$ | concatenation |
| $\delta$ | ReLU activation [51] |
| $\sigma$ | sigmoid activation |
| $\tanh$ | tanh activation |
| Softmax | softmax activation |
| BN | batch normalization [52] |
| Expand | expan input by repetition |

is shown in Fig. 1 and further explained in Fig. 2: it is based around data domain. Some methods consider the question of *when* the important data occurs, or others *where* it occurs, etc., and accordingly try to find key times or locations in the data. We divide existing attention methods into six categories which include four basic categories: channel attention (*what to pay attention to* [50]), spatial attention (*where to pay attention*), temporal attention (*when to pay attention*) and branch channel (*which to pay attention to*), along with two hybrid combined categories: channel & spatial attention and spatial & temporal attention. These ideas are further briefly summarized together with related works in Tab. 2.

The main contributions of this paper are:

- a systematic review of visual attention methods, covering the unified description of attention mechanisms, the development of visual attention mechanisms as well as current research,
- a categorisation grouping attention methods according to their data domain, allowing us to link visual attention methods independently of their particular application, and
- suggestions for future research in visual attention.

Sec. 2 considers related surveys, then Sec. 3 is the main body of our survey. Suggestions for future research are given in Sec. 4 and finally, we give conclusions in Sec. 5.

## 2 Other Surveys

In this section, we briefly compare this paper to various existing surveys which have reviewed attention methods and visual transformers. Chaudhari et al. [140] provide a survey of attention models in deep neural networks which concentrates on their application to natural language processing, while our work focuses on computer vision. Three more specific surveys [141], [142], [143] summarize the development of visual transformers while our paper reviews attention mechanisms in vision more generally, not just self-attention mechanisms. Wang et al. [144] present a survey of attention models in computer vision, but it only considers RNN-based attention models, which form just a part of our survey. In addition, unlike previous surveys, we provide a classification which groups various attention methods according to their data domain, rather than according to their field of application. Doing so allows us to concentrate

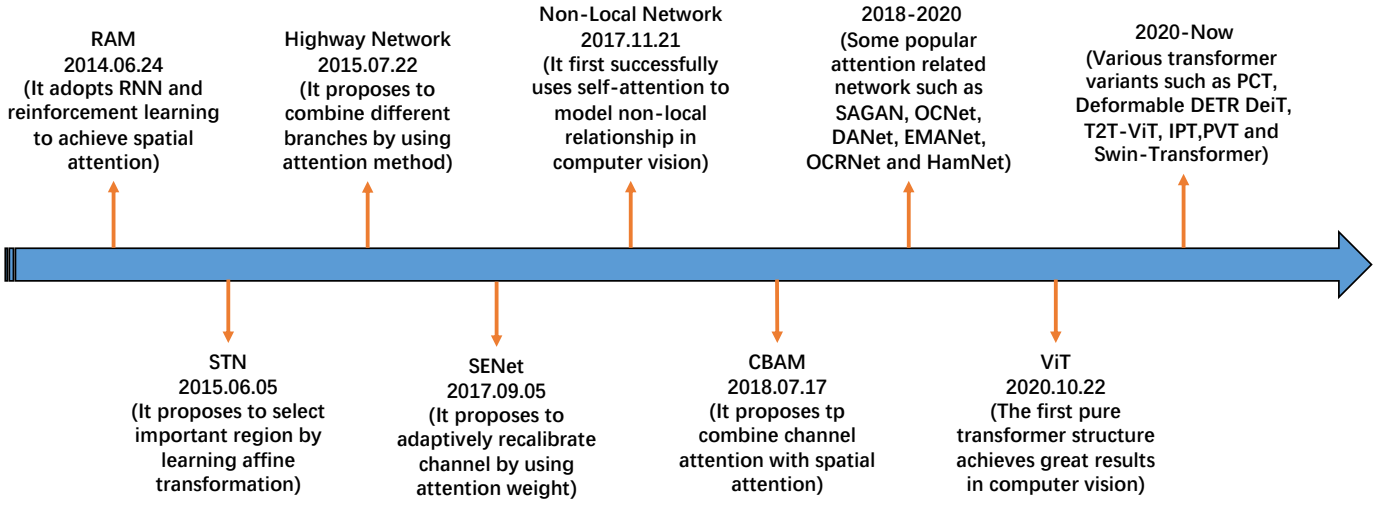


Fig. 3. Brief summary of key developments in attention in computer vision, which have loosely occurred in four phases. Phase 1 adopted RNNs to construct attention, a representative method being RAM [31]. Phase 2 explicitly predicted important regions, a representative method being STN [32]. Phase 3 implicitly completed the attention process, a representative method being SENet [5]. Phase 4 used self-attention methods [15], [33], [34].

TABLE 2
Brief summary of attention categories and key related works.

| Attention category | Description | Related work |
|---|---|---|
| Channel attention | Generate attention mask across the channel domain and use it to select important channels. | [5], [37], [53], [54], [55], [56], [57], [58] [25], [59], [60] |
| Spatial attention | Generate attention mask across spatial domains and use it to select important spatial regions (e.g. [15], [61]) or predict the most relevant spatial position directly (e.g. [7], [31]). | [8], [9], [15], [21], [31], [32], [34], [35] , [22], [26], [62], [63], [64], [65], [66], [67] , [41], [68], [69], [70], [71], [72], [73], [74] , [8], [34], [42], [43], [75], [76], [77], [78] , [27], [44], [45], [46], [79], [80], [81], [82] , [61], [83], [84], [85], [86], [87], [88], [89] , [47], [90], [91], [92], [93], [94], [95], [96] , [97], [98], [99], [100], [101], [102], [103], [104] , [20], [105], [106], [107], [108], [109] |
| Temporal attention | Generate attention mask in time and use it to select key frames. | [110], [111], [112] |
| Branch attention | Generate attention mask across the different branches and use it to select important branches. | [113], [114], [115], [116] |
| Channel & spatial attention | Predict channel and spatial attention masks separately (e.g. [6], [117]) or generate a joint 3-D channel, height, width attention mask directly (e.g. [118], [119]) and use it to select important features. | [6], [50], [117], [119], [120], [121], [122], [10], [101], [118], [123], [124], [125], [126] , [13], [14], [127], [128], [129] |
| Spatial & temporal attention | Compute temporal and spatial attention masks separately (e.g. [16], [130]), or produce a joint spatiotemporal attention mask (e.g. [131]), to focus on informative regions. | [130], [132], [133], [134], [135], [136], [137], [138], [139] |

on the attention methods in their own right, rather than treating them as supplementary to other tasks.

## 3 Attention Methods in Computer Vision

In this section, we first sum up a general form for the attention mechanism based on the recognition process of human visual system in Sec. 3.1. Then we review various categories of attention models given in Fig. 1, with a subsection dedicated to each category. In each, we tabularize representative works for that category. We also introduce that category of attention strategy more deeply, considering its development in terms of motivation, formulation and function.

### 3.1 General form

When seeing a scene in our daily life, we will focus on the discriminative regions, and process these regions quickly. The above process can be formulated as:

$$Attention = f(g(x), x) \tag{1}$$

Here $g(x)$ can represent to generate attention which corresponds to the process of attending to the discriminative regions. $f(g(x), x)$ means processing input $x$ based on the attention $g(x)$ which is consistent with processing critical regions and getting information.

With the above definition, we find that almost all existing attention mechanisms can be written into the above formulation. Here we take self-attention [15] and squeeze-and-